

Solution Architecture

The architecture separates the application into layers so each part can be built and improved independently.

Layer	Components	Role
Presentation	Web or mobile interface	Collects location input and displays results
Application	Backend service	Validates input, calls services, combines results
Data services	Weather API	Provides current conditions and forecast data
Intelligence	AI model	Creates the summary, suggestions and alert wording
Storage	Cache and saved preferences	Speeds up responses and remembers user choices

Flow

User > Interface > Backend > Weather API > AI model > Backend > Interface > User